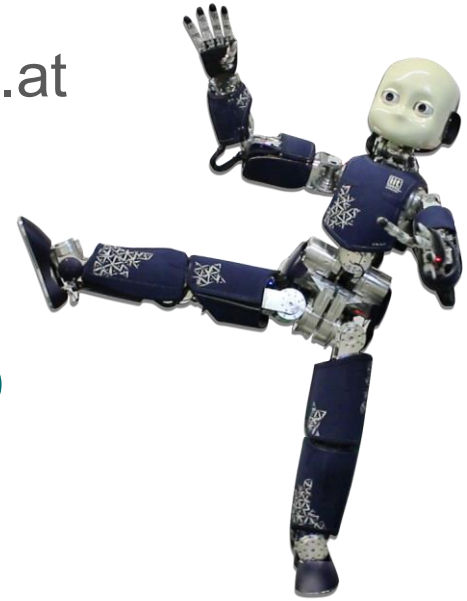
Project 1 - Steel Production Data

Julian Retzek

Email: julian-henry.retzek@stud.unileoben.ac.at



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Project Overview

Goal

- Predict a steel production quality parameter from various input parameters

Dataset

- CSV Table with 22 columns (output + input1–input21)
- All values already normalized (values between [0, 1]

Evaluation metrics

- R^2 (coefficient of determination)
- RSME (Root mean square error)

Model(s)

- Random Forrest Model
- XGBoost Modell

Sequence of Methods

Step 1 - Splitting the data

- Data was split 80/20 for training/testing with a fixed random seed

Step 2 - Creating and Training Random Forrest Model

- Random Forrest Model builds 100 decision trees that each learn a subset of the training data. Final prediction is average of all the 100 individual tree predictions.

Step 3 – Writing a test class

- After evaluating the R^2 (which was only 47%) revised model multiple times, but the value stayed low.

Step 4 - XGBoost as an alternative

- Creating an alternative, more precise model. XGBoost builds the 500 trees in sequential order and tries to correct the former inputs. However, this did also not yield different results

Step 5 - Investigating of the low R^2 Value

- SD is very low. Since Values vary so little, small absolute prediction errors appear large, which brings R^2 down.
- Better to look at RSME, which is <0.1 → prediction is (in absolute terms) close to true value of output

Results

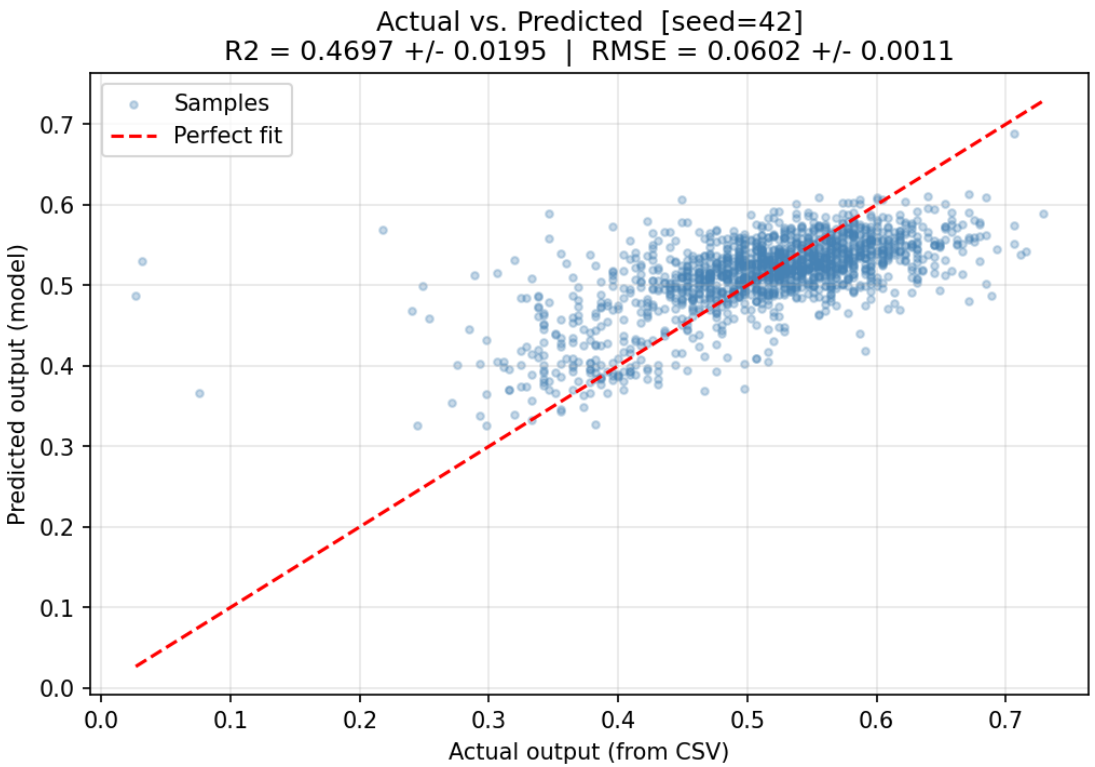
Model performance on test set

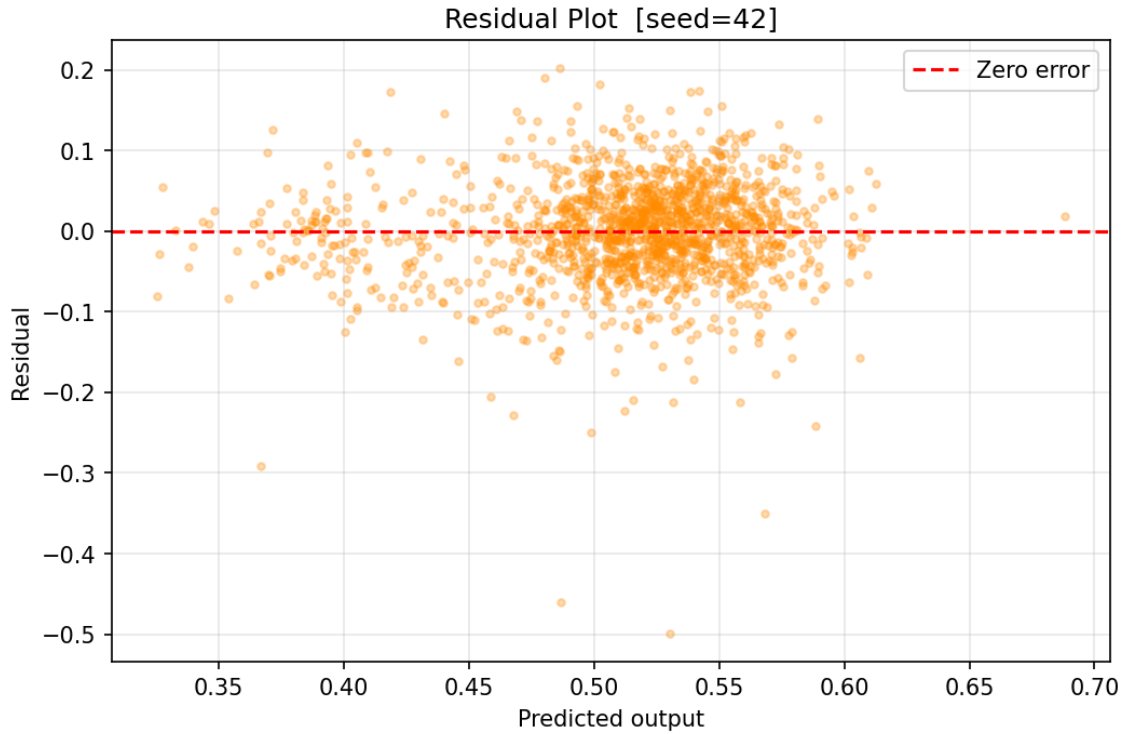
Model	R^2 Score +- SD	RMSE	No of Runs
Random Forrest	0.4697 ± 0.0195	0.0602 ± 0.0011	10
XGBoost	0.4852 ± 0.0239	0.0593 ± 0.0012	10

- R^2 = 0,47 → Model only predicts 47% of quality difference correctly
- Low R^2 is caused by small output variance, not by poor model
- RMSE <0,1 means predictions are numerically close to true values → R^2 can be lower

Visualisation

Visualisation for Random Forrest Model





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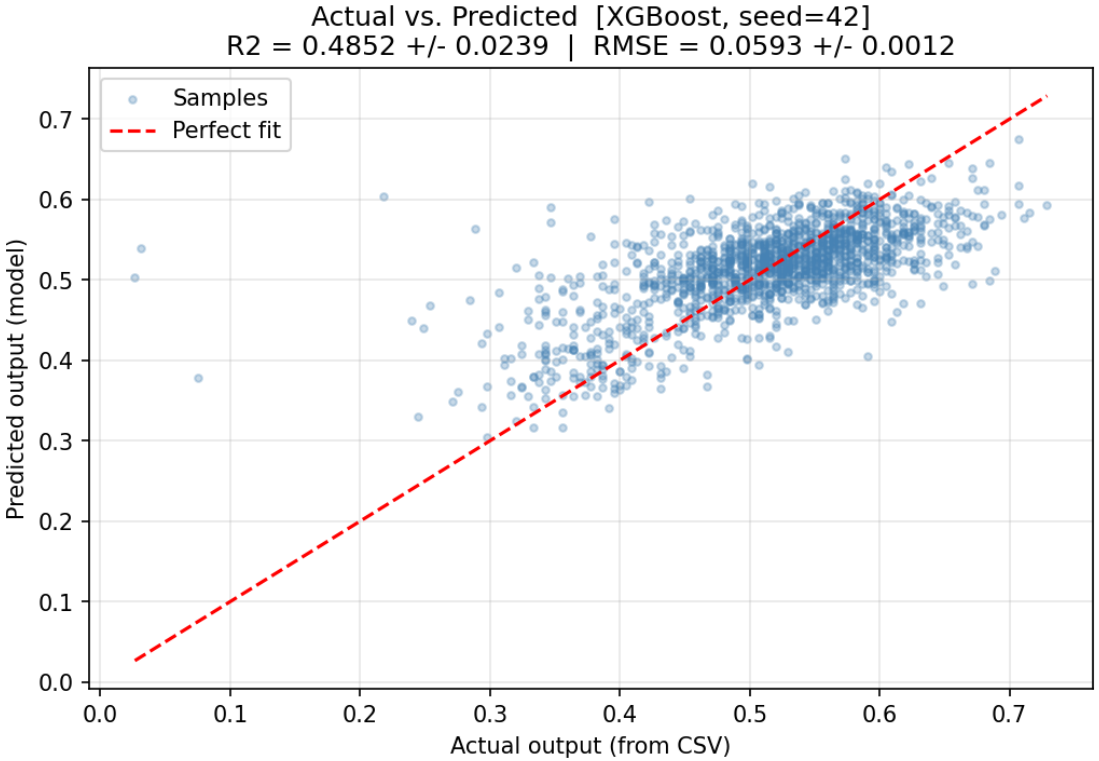
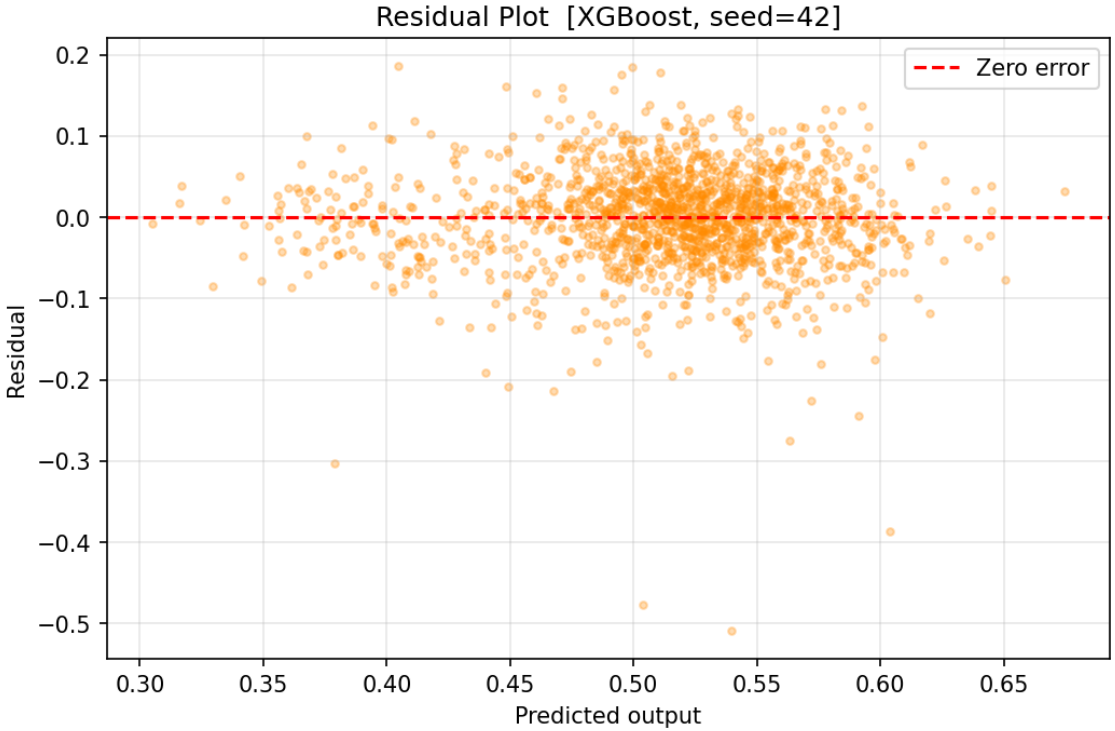
Visualisation

Visualisation for XGBoost Model





Conclusion

The model(s) gives a good prediction

Lessons learned:

- Not one but two models were built because the R^2 parameter seemed too low → time and effort was spent (in vain) because of focusing on the wrong metric instead of properly **looking at the data**

Future Outlook

- Model could be further improved by
  - outlier analysis of input parameters
  - k-fold cross-validation instead of randomsplit of data

Thank you for your attention!

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